FRANCK PORTIER
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M1 MACRO
GROUP A
Session 18

These notes are written during class. They are posted "for the record" and may contain mistakes. They are not intended to be read by anyone that did not attend the class.

$$y_t = f(x_{1t}, x_{2t})$$

$$\hookrightarrow \quad \boxed{\hat{y}_t = \frac{\bar{x}_1 f_1}{f}\,\hat{x}_{1t} + \frac{\bar{x}_2 f_2}{f}\,\hat{x}_{2t}}$$

$$f_1 = f'_1(\bar{x}_1, \bar{x}_2)$$

$$f_2 = f'_2(\bar{x}_1, \bar{x}_2)$$

$$, \; f = f(\bar{x}_1, \bar{x}_2) = \bar{y}$$

$$\hat{x}_t = \frac{x_t - \bar{x}}{\bar{x}}$$

Examples of linearization

(1) $\underline{y_t = z_t h_t^\alpha} \Rightarrow y_t = f(z_t, h_t)$

$\underline{SS}$ $\quad \bar{y} = \bar{z}\,\bar{h}_t^\alpha$

$$\text{(1)} \Rightarrow \quad \hat{y}_t = \frac{\bar{z} \times \overbrace{f_1}^{\bar{h}^\alpha}}{\underbrace{f}_{\bar{z}\bar{h}^\alpha}}\,\hat{z}_t + \frac{\bar{h} \times \overbrace{f_2}^{\bar{z}\alpha\bar{h}^{\alpha-1}}}{\underbrace{f}_{\bar{z}\bar{h}^\alpha}}\,\hat{h}_t$$

$$\boxed{\hat{y}_t = \hat{z}_t + \alpha \hat{h}_t}$$

$\rightarrow$ linearized version of eq (1)

(note $\hat{x}_t \simeq \log x_t - \log \bar{x}$)

(2) $y_t = c_t + i_t + g_t = f(c_t, i_t, g_t)$

<u>Linearization</u>

$$\hat{y}_t = \frac{\bar{c}\,\overbrace{f_1}^{1}}{\underbrace{f}_{\bar{y}}}\,\hat{c}_t + \frac{\bar{i}\,\overbrace{f_2}^{1}}{\underbrace{f}_{\bar{y}}}\,\hat{i}_t + \frac{\bar{g}\,\overbrace{f_3}^{1}}{\underbrace{f}_{\bar{y}}}\,\hat{g}_t$$

$$\boxed{\hat{y}_t = \frac{\bar{c}}{\bar{y}}\,\hat{c}_t + \frac{\bar{i}}{\bar{y}}\,\hat{i}_t + \frac{\bar{g}}{\bar{y}}\,\hat{g}_t}$$

Back to the habit formation model.

$$\begin{cases} \dfrac{1}{c_t - b\,c_{t-1}} - E_t \dfrac{\beta b}{c_{t+1} - b\,c_t} = \dfrac{1}{z_t} \\ x_t = \dfrac{1}{c_t - b\,c_{t-1}} \end{cases}$$

SS

$$\bar{x} = \frac{1}{(1-b)\,\bar{c}} \longrightarrow \boxed{\bar{c} = \frac{(1-\beta b)}{1-b}\,\bar{z}}$$

$$\bar{x} - \beta b\,\bar{x} = \frac{1}{\bar{z}} \Rightarrow \boxed{\bar{x} = \frac{1}{(1-\beta b)\,\bar{z}}}$$

$$X_t - \beta b E_t X_{t+1} = \frac{1}{z_t}$$

$$f(X_t, X_{t+1}) = g(z_t)$$

$$\left(f(\bar{x}, \bar{x}) = g(\bar{z}) \right)$$

↳ linearization

$1 \qquad \beta b \qquad -\frac{1}{z^2}$

$$\frac{\bar{x} f_1}{f} \hat{x}_t - \frac{\bar{x} f_2}{f} E_t \hat{x}_{t+1} = \frac{\bar{z} g'}{g} \hat{z}_t$$

$\frac{1}{z}$

$(1-\beta b)\bar{x} = \frac{1}{z}$

$$\frac{\bar{x}}{(1-\beta b)\bar{x}} \hat{x}_t - \frac{\beta b \bar{x}}{(1-\beta b)\bar{x}} E_t \hat{x}_{t+1} = -\bar{z} \times \frac{1}{\bar{z}^2} \times \bar{z} \times \hat{z}_t$$

$$\frac{1}{1-\beta b}\hat{x}_t - \frac{\beta b}{(1-\beta b)} E_t \hat{x}_{t+1} = -\hat{z}_t$$

$$\Leftrightarrow \boxed{\hat{x}_t = \beta b E_t \hat{x}_{t+1} - (1-\beta b)\hat{z}_t}$$

Solving :

$$\hat{x}_t = -(1-\beta b)\hat{z}_t + \beta b E_t\left[\beta b E_{t+1}\hat{x}_{t+2} - (1-\beta b)\hat{z}_{t+1}\right]$$

$$= -(1-\beta b)\hat{z}_t - \beta b \times (1-\beta b) E_t \hat{z}_{t+1} + (\beta b)^2 E_t \hat{x}_{t+2}$$

$$\hat{X}_t = -(1-\beta b) \times \sum_{i=0}^{\infty} \underbrace{(\beta b)}_{\in]0,1[}{}^{i} E_t Z_{t+i}$$

$$+ \underbrace{\lim_{j \to \infty} (\beta b)^j E_t \hat{X}_{t+j}}_{= 0 \text{ (TRANSVERSALITY)}}$$

Assume $Z_t = (1-\rho) B + \rho Z_{t-1} + \varepsilon_t$

SS: $\bar{Z} = (1-\rho) B + \rho \bar{Z} \Rightarrow \bar{Z} = B$

$$z_1 = (1-\rho)\bar{z} + \rho z_{t-1} + \varepsilon_t$$

$$\frac{z_t - \bar{z}}{\bar{z}} = \rho\,\frac{(z_{t-1} - \bar{z})}{\bar{z}} + \frac{\varepsilon_t}{\bar{z}}$$

$$\underline{\hat{z}_t = \rho\,\hat{z}_{t-1} + \tilde{\varepsilon}_t}\;, \quad 0 < \rho < 1$$

↳

$$E_t(\hat{z}_{t+1}) = \rho$$

$$E_t(\hat{z}_{t+2}) = \rho^2$$

$$\vdots$$

$$E_t[\hat{z}_{t+i}] = \rho^i$$

Then

$$\hat{x}_t = -(1-\beta b) \sum_{i=0}^{\infty} (\beta b)^i E_t[z_{t+i}]$$

$$= -(1-\beta b) \sum_{i=0}^{\infty} (\beta b)^i \rho^i z_t$$

$$= -(1-\beta b)\, z_t \times \underbrace{\sum_{i=0}^{\infty} \underbrace{(\beta b \rho)}_{\in]0,1[}{}^i}_{\frac{1}{1-\beta b \rho}}$$

Solution

$$\boxed{\hat{x}_t = -\frac{(1-\beta b)}{1-\beta b \rho} \times z_t}$$

$\longrightarrow$ what about q?

$$x_t = \frac{1}{c_t - b\,c_{t-1}} = f(c_t, c_{t-1})$$

↳ linearization

$$-\frac{1}{(\bar c - b\bar c)^2} \qquad \frac{b}{(\bar c - b\bar c)^2}$$

$$\hat x_t = \frac{\bar c\, f_1}{f}\,\hat c_t + \frac{\bar c\, f_2}{f}\,\hat c_{t-1}$$

$, f$

$$+\ \bar x = \frac{1}{(1-b)\bar c} \qquad \frac{1}{(1-b)\bar c}$$

$$\hat x_t = \bar c \times \frac{-1}{\bar c^2 (1-b)^2} \times (1-b)\bar c \times \hat c_t + \bar c \times \frac{b}{\bar c^2 (1-b)^2} \times (1-b)\bar c\ \hat c_{t-1}$$

$$\hat x_t = -\frac{1}{1-b}\,\hat c_t + \frac{b}{1-b}\,\hat c_{t-1} \Rightarrow \boxed{\hat c_t = b\,\hat c_{t-1} - (1-b)\,\hat x_t}$$

$$\hat{c}_t = b\,\hat{c}_{t-1} + \frac{(1-b)(1-\beta b)}{1-\beta b \rho}\,\hat{z}_t$$

let $L\hat{c}_t = \hat{c}_{t-1}$

$$\hat{c}_t = bL\,\hat{c}_t + \frac{(1-b)(1-\beta b)}{1-\beta b \rho} \times \frac{\tilde{\varepsilon}_t}{1-\rho L}$$

$$\left[\begin{aligned} \hat{z}_t = \rho\,\hat{z}_{t-1} + \tilde{\varepsilon}_t \iff \hat{z}_t &= \rho L \hat{z}_t + \tilde{\varepsilon}_t \\ \iff (1-\rho L)\,\hat{z}_t &= \tilde{\varepsilon}_t \\ \hat{z}_t &= \frac{\tilde{\varepsilon}_t}{1-\rho L} \end{aligned}\right]$$

$$\boxed{\hat{c}_t (1-bL)(1-\rho L) = \frac{(1-\beta b)(1-b)}{1-\beta b \rho}\tilde{\varepsilon}_t}$$

= solution

$$(1-bL)(1-\rho L) = 1-(\rho+b)L+\rho b L^2$$

$$\boxed{\hat{c}_t = (\rho+b)\hat{c}_{t-1} - \rho b\, \hat{c}_{t-2} + \frac{(1-\beta b)(1-b)}{1-\beta b \rho}\tilde{\varepsilon}_t}$$

SOLUTION

↳ AR(2)

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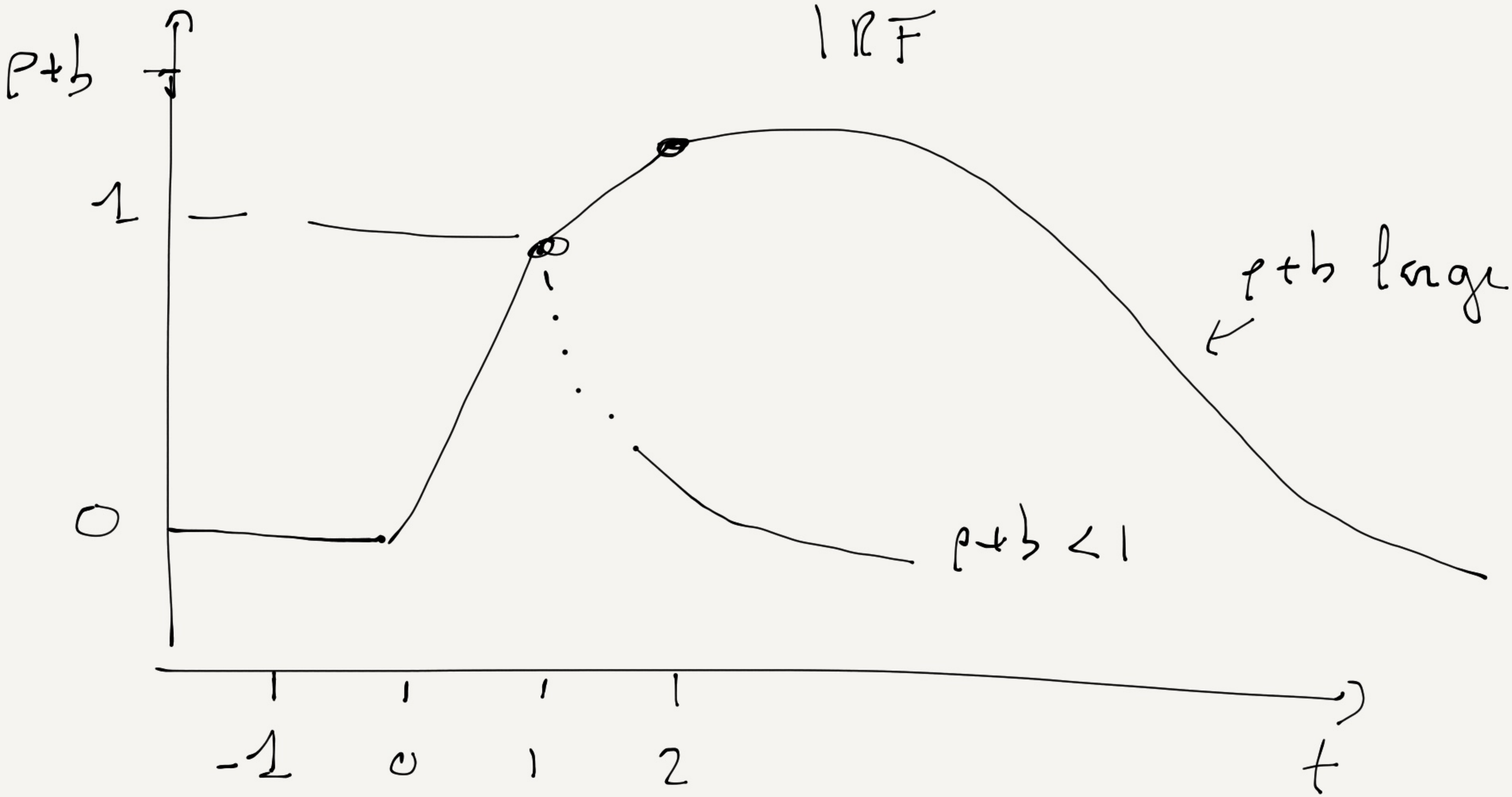

$$\tilde{c}_1 = 1 \quad , \quad \tilde{c}_2 = \rho + b \quad , \quad \tilde{c}_3 = (\rho+b)^2 - \rho b$$